6.6 Solving Exponential and Logarithmic Equations

Name _____ Period ____ Date _____

Algebra II 6.6 · Student sheet

I can…

- solve exponential equations.
- solve logarithmic equations and check for extraneous solutions.
- solve exponential inequalities and real-life problems.

Vocabulary · fill in the missing word

Property of Equality for Exponential Equations — b^x = _____ = y

Property of Equality for Logarithmic Equations — $\log_b x$ = _____ = y

Extraneous solution — an answer that makes the log of a _____ number in the original equation

Part 1 · Same base or take a log?

Sort each equation. Then solve one equation from each column.

$2^x = 16$ $9^x = 27$ $5^x = 12$ $e^x = 4$ $4^x = 1/64$ $10^x = 7$

Rewrite with the same base	Take a log of both sides

Part 2 · Notes: fill in the blanks

If $b^x = b^y$, then _____.

If the bases can't match, take a _____ of both sides: $b^x = a \rightarrow x =$ _____.

To solve $\log_b x = c$, rewrite it as _____.

Check every answer: the inside of a logarithm must be _____.

Part 3 · Examples (work them with the class)

Example 1

Solve $9^x = 27^{x-1}$.

Example 2

Solve $4^x = 11$. Round to three decimal places.

Example 3

Solve $\ln(4x - 7) = \ln(x + 5)$.

Example 4

Solve $\log_2(5x - 17) = 3$.

Example 5

Solve $\log_2 x + \log_2(x - 7) = 3$.

Example 6

Solve $3^x < 20$. Round to three decimal places.

Example 7

You deposit $1,000 at 6% annual interest compounded continuously. How long until the money doubles? Round to two decimal places.

Part 4 · Partner practice

Partner 1

Solve $5^x = 30$. Round to three decimal places.

Partner 2

Solve $\log_4(x + 3) = 2$.